

Convergent and restricted power series

Notation 1. Let $T = (T_1, \dots, T_n)$ be a tuple of n indeterminates, $r = (r_1, \dots, r_n)$ be a tuple of n positive real numbers, and $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ be a multi-index. We use the following notations:

- $T^\alpha := T_1^{\alpha_1} T_2^{\alpha_2} \dots T_n^{\alpha_n}$ and $r^\alpha := r_1^{\alpha_1} r_2^{\alpha_2} \dots r_n^{\alpha_n}$;
- $\underline{T/r} := (T_1/r_1, T_2/r_2, \dots, T_n/r_n)$;
- $|\alpha| := \alpha_1 + \alpha_2 + \dots + \alpha_n$;

1 Absolutely convergent power series

Definition 2. Let R be a banach ring and $r > 0$ be a real number. We define the *ring of absolutely convergent power series* over R with radius r as

$$R \langle T/r \rangle := \left\{ \sum_{n=0}^{\infty} a_n T^n \in R[[T]] : \sum_{n=0}^{\infty} |a_n| r^n < \infty \right\}.$$

Equipped with the norm $\| \sum_{n=0}^{\infty} a_n T^n \| := \sum_{n=0}^{\infty} |a_n| r^n$, the ring $R \langle T/r \rangle$ is a banach ring.

For a tuple of n indeterminates $T = (T_1, \dots, T_n)$ and a tuple of n positive real numbers $r = (r_1, \dots, r_n)$, we define

$$R \langle \underline{T/r} \rangle := R \langle T_1/r_1, \dots, T_n/r_n \rangle := R \langle T_1/r, \dots, T_{n-1}/r_{n-1} \rangle \langle T_n/r_n \rangle.$$

Note that if R has trivial norm, then

$$R \langle T/r \rangle = \begin{cases} R[[T]], & \text{if } r < 1; \\ R[T], & \text{if } r \geq 1. \end{cases}$$

Proposition 3. Let R be a banach ring and $r > 0$ be a real number. Then the ring of absolutely convergent power series $R \langle T/r \rangle$ is a banach ring. **Yang: To be checked.**

In general, the ring of absolutely convergent power series is not even power-multiplicative. The following examples describe the uniformization of the ring of absolutely convergent power series over fields.

Example 4. Let $A = \mathbb{C} \langle T/r \rangle$. For $f \in A$, set

$$\|f\|_0 := \max_{x \in E(0,r)} |f(x)|.$$

I claim that $\| \cdot \|_0 = \rho_A(\cdot)$. **Yang: To be continued.**

Example 5. Let $\mathbb{k}$ be an algebraically closed non-archimedean complete field and $A = \mathbb{k} \langle T/r \rangle$.

Yang: To be continued.

2 Tate algebras

Definition 6. Let R be a non-archimedean banach ring. The *Tate algebra* over R with radius $r > 0$ in one variable is defined as

$$R\{T/r\} := \left\{ \sum_{n=0}^{\infty} a_n T^n \in R[[T]] : |a_n| r^n \rightarrow 0 \text{ as } n \rightarrow +\infty \right\}.$$

For a tuple of n indeterminates $T = (T_1, \dots, T_n)$ and a tuple of n positive real numbers $r = (r_1, \dots, r_n)$, we define

$$R\{\underline{T}/r\} := R\{T_1/r_1, \dots, T_n/r_n\} := R\{T_1/r_1, \dots, T_{n-1}/r_{n-1}\}\{T_n/r_n\}.$$

Elements in $R\{\underline{T}/r\}$ are also called *restricted power series*.

Proposition 7. Let R be a non-archimedean banach ring. Then the Tate algebra $R\{T/r\}$ is a non-archimedean multiplicative banach R -algebra with respect to the *gauss norm*

$$\left\| \sum_{n \geq 0} a_n T^n \right\| := \sup_{n \geq 0} |a_n| r^n = \max_{n \geq 0} |a_n| r^n.$$

Proof. Yang: To be added. □

The following examples describe the reduction rings of Tate algebras.

Example 8. Let R be a non-archimedean banach ring and $A = R\{T\}$ be the Tate algebra in one variable over R . Then we have

$$A^\circ = \left\{ \sum_{n \geq 0} a_n T^n : |a_n| \leq 1 \text{ for all } n \in \mathbb{N} \right\} = R^\circ\{T\},$$

and

$$A^{\circ\circ} = \left\{ \sum_{n \geq 0} a_n T^n : |a_n| < 1 \text{ for all } n \in \mathbb{N} \right\} = R^{\circ\circ}\{T\}.$$

Since the norm of items in a restricted power series will tend to 0, we have

$$\tilde{A} = \tilde{R}[T].$$

Example 9. Let R is a multiplicative non-archimedean banach ring. Set

$$\sqrt{|R|^{-1}} = \{r \in \mathbb{R}_{>0} : r^{-n} \in |R| \text{ for some } n \in \mathbb{N}_{>0}\}.$$

Fix $r \in \mathbb{R}_{>0}^n$, consider the Tate algebra $A = R\{T/r\}$.

Suppose that $r \in \sqrt{|R|^{-1}}$. Let n be the minimal positive integer such that $r^n \in |R|^{-1}$ and

$$\tilde{M}_k := \{a \in R : |a| = r^{-nk}\} / \{a \in R : |a| < r^{-nk}\}.$$

For $a_m T^m$ with $n \nmid m$, we have $\|a_m T^m\| = |a_m| r^m \leq 1 \implies |a_m| r^m < 1$. Hence

$$\widetilde{R\{T/r\}} = \tilde{R} \oplus \tilde{M}_1 T^n \oplus \tilde{M}_2 T^{2n} \oplus \tilde{M}_3 T^{3n} \oplus \dots$$

In case $R = \mathbf{k}$ is a non-archimedean field, we have $\tilde{M}_k \cong \tilde{\mathbf{k}}$ by choosing an element $c \in \mathbf{k}$ with

$|c| = r^{-n}$. Hence

$$\widehat{\mathbf{k}\{T/r\}} \cong \widehat{\mathbf{k}[T^n]}.$$

Suppose that $r \notin \sqrt{|R|^{-1}}$. Then for every $\|a_n T^n\| = a_n r^n \leq 1$, we have $|a_n| < 1$. It follows that

$$\widehat{R\{T/r\}} = \widetilde{R}.$$

Definition 10. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. For $f = \sum_{a_n \in \mathbb{N}} a_n T^n \in A$, we define the *degree* of f as

$$\deg f := \max\{n \in \mathbb{N} : \|a_n\| r^n = \|f\|\}.$$

Easily to see that

$$\deg(f \cdot g) = \deg f + \deg g$$

and

$$\deg(f + g) \leq \max\{\deg f, \deg g\}.$$

It is interesting to note that if R has trivial norm, then $\deg f$ coincides with the usual degree of polynomials when $r \geq 1$ and the order of formal power series when $r < 1$.

Definition 11. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. A restricted power series $f = \sum_{n \in \mathbb{N}} a_n T^n \in A$ of degree d is said to be *distinguished* if a_d is invertible in R .

Proposition 12. Let R be a non-archimedean banach ring. An element $f \in R\{T/r\}$ is invertible if and only if it is distinguished of degree 0.

Proof. Denote $f = \sum_{n=0}^{\infty} a_n T^n$. Multiplying by a_0^{-1} , we can reduce to the case $a_0 = 1$. Let $g = \sum_{n=0}^{\infty} b_n T^n$ be the inverse of f in $R[[T]]$. Then we have

$$f \cdot g = \sum_{k=0}^{\infty} \left(\sum_{n+m=k} a_n b_m \right) T^k = 1 \implies \forall k \geq 1, b_k = - \sum_{\substack{n+m=k \\ n \neq 0}} a_n b_m.$$

Set $A = \|f - 1\| < 1$. We show that for every $m \in \mathbb{N}$, there exists $B_m > 0$ such that for all $k \geq B_m$, we have $\|b_k\| r^k \leq A^m$. This implies that $\|b_k\| r^k \rightarrow 0$ as $k \rightarrow +\infty$ and hence $g \in R\{T/r\}$.

Note that $b_0 = 1$. By induction on k , we have

$$\|b_k\| r^k \leq \max_{\substack{n+m=k \\ n \neq 0}} \|a_n\| r^n \cdot \|b_m\| r^m \leq A \max_{m < k} \|b_m\| r^m \leq A.$$

Suppose that the claim holds for m . There exists $C_{m+1} \in \mathbb{N}$ such that for all $n \geq C_{m+1}$, we have $\|a_n\| r^n \leq A^{m+1}$. Set $B_{m+1} = B_m + C_{m+1} + 1$. For any $k \geq B_{m+1}$, we have

$$\|b_k\| r^k \leq \max_{\substack{n+m=k \\ n \neq 0}} \|a_n\| r^n \cdot \|b_m\| r^m \leq \max\{A^{m+1}, A \cdot A^m\} = A^{m+1}$$

since either $n \geq C_{m+1}$ or $m \geq B_m$. Hence the claim holds for $m + 1$. $\square$

3 Weierstrass preparation

Definition 13. Let R be a non-archimedean banach ring and $A = R\{T/r\}$. A Weierstrass polynomial is a monic polynomial $P \in A[T] \subset R\{T/r\}$ whose two degrees as a polynomial and as a restricted power series coincide.

Theorem 14 (Weierstrass preparation theorem). Let R be a non-archimedean banach ring. Let $f \in R\{T/r\}$ be a distinguished restricted power series of degree d . Then there exists a unique Weierstrass polynomial $p \in R[T]$ of degree d and a unique unit $u \in R\{T/r\}$ such that

$$f = p \cdot u.$$

Yang: To be checked.

Proof. Yang: To be added. □

Remark 15. In my knowledge, there are at least three different versions of Weierstrass preparation theorem under different settings:

- The classical Weierstrass preparation in complex analysis;
- The Weierstrass preparation for formal power series over complete noetherian local rings;
- The Weierstrass preparation for Tate algebras over non-archimedean banach rings.

Let $(R, \mathfrak{m})$ be a complete noetherian local ring. Note that there is also a Weierstrass preparation theorem for formal power series over R stating that for every formal power series $f \in R[[T]]$ whose reduction $\bar{f} \in (R/\mathfrak{m})[[T]]$ is of order d , there exists a unique monic polynomial $p \in R[T]$ of degree d and a unique unit $u \in R[[T]]$ such that

$$p \equiv T^d \pmod{\mathfrak{m}}, \quad f = p \cdot u.$$

Yang: To be continued.

Corollary 16. Let $\mathbf{k}$ be a complete non-archimedean field and $r > 0$ be a positive real number. Then the Tate algebra $\mathbf{k}\{T/r\}$ is an euclidean domain with respect to the degree defined in [Definition 10](#).

Yang: To be added.

Proof. Let $f, g \in \mathbf{k}\{T/r\}$ be two elements with $g \neq 0$. We need to find $q, s \in \mathbf{k}\{T/r\}$ such that

$$f = q \cdot g + s, \quad \deg s < \deg g.$$

Yang: To be added. □

Appendix